

CoRoT-2b

R-0227

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_240828 · created 2026-07-17T02:50:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460550.6773 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.039 +/- 0.06
Transit depth (Rp/Rs)^2	0.16 +/- 0.47 [%]
Orbital Inclination (inc)	85.0 +/- 2.03
Airmass coefficient 1 (a1)	0.376 +/- 0.021
Airmass coefficient 2 (a2)	0.81 +/- 0.38
Scatter in the residuals of the lightcurve fit is	5.75 %
Best Comparison Star	#2 - [271, 408]
Optimal Aperture	2.44
Optimal Annulus	16.84
Transit Duration (day)	0.057 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.039 $\pm$ 0.06 vs 0.1667 (deviation 2.13 σ)
Duration fit vs published	0.057 d vs 0.095 d (deviation 2.71 σ)
Mid-time O-C	8.2 min
Depth SNR	0.7
Residual scatter	5.75 %

Light curve

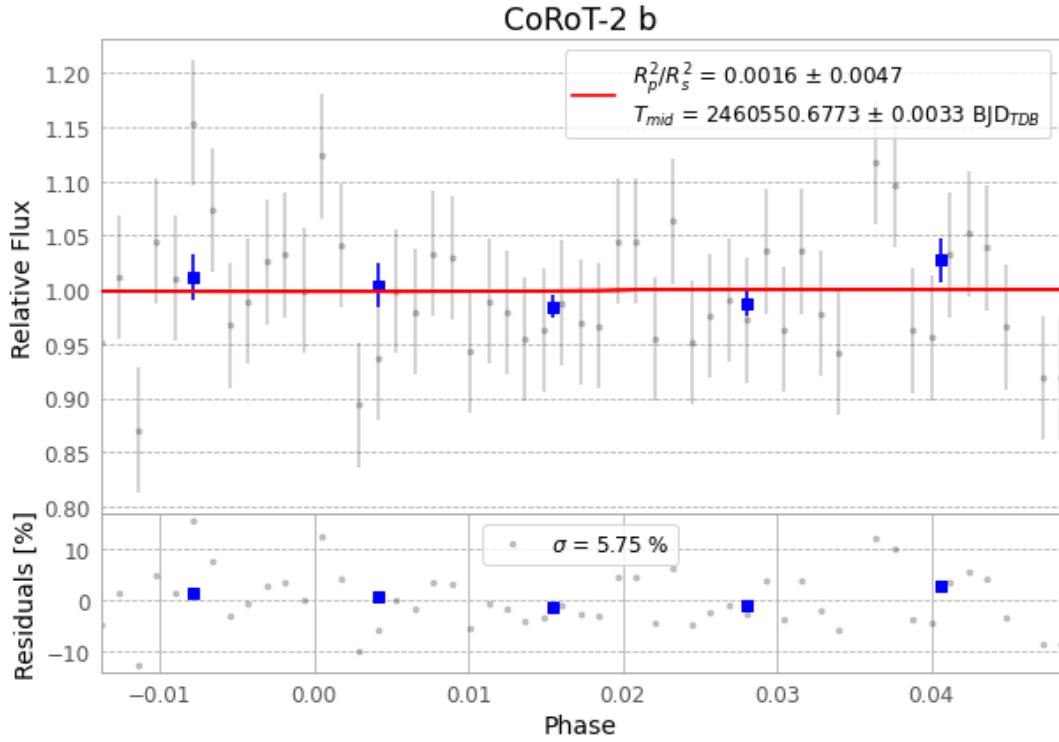


Figure 1 — FinalLightCurve_CoRoT-2 b_2024-08-27.png (SHA-256 f99cc23f007a741c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1e4e1ce6757f5d42...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

56 FITS frames (37 MB), CoRoT-2240828032415.FITS → CoRoT-2240828060915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-240828.log (d36dc2ab206f...), FinalLightCurve_CoRoT-2 b_2024-08-27.png (f99cc23f007a...), FinalLightCurve_CoRoT-2 b_2024-08-27.csv (a64c1b7fbd87...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 02:50Z.